

Hardware-aware in-situ learning in an analog spiking neural network.*

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Abstract

Hardware-aware in-situ learning in an analog spiking neural network using measured memristive synapses and neurons, with a compact device model and a small fabricated demonstrator.

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1 Introduction

The goal of the summer project is to identify a reasonable scope for the PhD research. This report covers an introduction, literature review, proposed methodology, impact assessment, plan of future work, and references.

1.1 Context

The human brain is often quoted for its extraordinary efficiency, with consumption of 20w compared with the kilo- or mega-watts of power needed to operate a traditional computer of equivalent capability, where data transfers between CPU and external memory can consume approximately 10-100x more energy than the logic operation itself (Khan et al. 2024). In-memory computing (IMC) reduces this Von-Neumann architecture cost by performing computation within to the memory array, thereby minimizing data movement.

Neuromorphic computers have the potential to perform complex calculations more power-efficiently and on a smaller footprint than digital Von-Neumann systems. The term “neuromorphic” was created by Carver Mead in 1990 (Mead 1990), and early research was on faster computation with neural networks outperforming traditional Von-Neumann architectures, exploiting natural parallelism and custom hardware to complete neural-style computations. This was used for real-time control, real-time digital image reconstruction, and autonomous robot control. Focus later moved to power consumption and efficiency.

A key differentiator is the level of biological plausibility: whether biologically-plausible (e.g. Hodgkin-Huxley or Morris Lecar neuron models), biologically-inspired (replicating behavior of neural systems but not in a biologically-plausible way, e.g. Izhikevich), or artificial neuron models such as Leaky Integrate-and-Fire (LIF) and McCulloch-Pitts (e.g. sigmoid, tanh activation functions) (Schuman et al. 2017).

1.2 Motivation and Significance

The Internet of Things (IOT) requires intelligent information processing in the field, away from powerful and energy-consuming cloud and datacentre-based AI capabilities. Without low-energy local AI inference, very large amounts of data need to be transmitted for central processing, requiring energy, reliable bandwidth and connectivity, latency, and the provision of land, energy grid and water supplies (Jiang et al. 2020). IOT devices often have a limited energy budget, constrained by energy harvesting and battery capacity. Any lack of energy self-sufficiency requires ongoing human involvement in battery maintenance and replacement.

Much current machine learning research assumes more powerful models running on centralised hardware, with the primary aim of improving accuracy against well-established benchmark datasets and problems (Thomas and Uminsky 2022). Energy demands of frontier models (especially for training) are rising rapidly and will become the

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limiting factor for further adoption; datacentres cannot be built because of the lack of energy grid and cooling water availability (Chen et al. 2026).

In contrast, neuromorphic computing seeks to emulate the brain’s event-driven, massively parallel processing using energy-efficient hardware models such as spiking neural networks (SNNs). Maass’s 1997 paper (“Networks of spiking neurons”) is cited as the conceptual foundation for SNNs, and argues that a biologically relevant function which requires hundreds of hidden units on a sigmoidal artificial neural network (ANN) could be computed by a single spiking neuron (Maass 1997).

Memristor device architectures (Strukov et al. 2008) show promise as implementations with bio-plausible behaviour, because drift memristors naturally implement analog synaptic weights and non-volatile (long-term) plasticity, while diffusive memristors provides short term neuron firing behaviour (Hu et al. 2022).

Recent work has demonstrated memristor CMOS neuromorphic devices with spike-timing-dependent-plasticity (STDP) learning at the synapse level (Bianchi et al. 2020), and small-scale SNNs with in-situ adaptation (Z. Wang et al. 2018), but most demonstrations either remain shallow (Covi et al. 2016), rely on ex-situ weight programming (Wijesinghe et al. 2018), or require tight integration with digital neuromorphic platforms such as FPGAs (Javanshir et al. 2022).

Power consumption for inference in extremely constrained environments is measured, but often only for single devices (synapses, neurons) (Chauhan et al. 2025), not for complete systems (Zhang et al. 2024). This project aims to bridge these developments by designing, fabricating, and evaluating a multilayer memristor-based SNN with in-situ learning. The focus will be on time-series tasks such as event-based vision.

2 Literature Review

This literature review surveys recent work at the intersection of analog spiking neural networks (SNNs), memristive and ferroelectric/antiferroelectric devices, and hardware-aware in-situ learning, with a focus on compact device models and fabricated demonstrators.

2.1 Search Strategy

A literature search was performed using IEEE Xplore and Arxiv, based on publications and conference submissions from 2010 to 2026. The following keywords were used:

Topic focus:

- “spiking neural network” OR “SNN”
- “neuromorphic computing” OR “neuromorphic hardware”
- “in-situ learning” OR “on-chip learning” OR “hardware learning”
- “memristor” OR “memristive synapse” OR “crossbar array”
- “ReRAM” OR “ferroelectric” OR “antiferroelectric”

Learning rules and algorithms:

- “STDP” OR “spike-timing-dependent plasticity”
- “VDSP” OR “voltage-dependent synapse plasticity”
- “three-factor learning” OR “eligibility trace” OR “e-prop”
- “Hebbian learning” OR “Oja’s rule” OR “surrogate gradient” for SNN training

Neuron and synapse models:

- “leaky integrate-and-fire” OR “LIF”
- “Izhikevich” OR “Hodgkin-Huxley”
- “short-term plasticity” OR “long-term plasticity” OR “STP” OR “LTP”
- “potentiation” OR “depression”

2.2 Surveys and Contextual Reviews

Rajendran et al. determine the operating characteristics of novel nanoscale devices that could be used to fabricate neuromorphic systems and show that neuromorphic systems based on nanoscale devices can potentially improve

density and power consumption by at least a factor of 10, compared with conventional CMOS (Rajendran et al. 2013). Rajendran et al. also review emerging memory technologies and how they can enable energy-efficient implementation of large neuromorphic systems. They highlight some of the key aspects of biological computation that are being mimicked in these novel nanoscale devices, and discuss various strategies employed to implement them efficiently (Rajendran and Alibart 2016).

Schuman et al.’s survey of neuromorphic computing provides a longer historical overview spanning neuro-inspired models, learning approaches, hardware/devices, supporting systems, and applications, showing how memristive and ferroelectric neuromorphic hardware fits into the broader landscape (Schuman et al. 2017). A subsequent article on opportunities for neuromorphic algorithms and applications by the same group highlights open challenges in algorithm-hardware co-design, robustness, and application-driven benchmarks. It shows a lack of readily accessible and usable software and hardware systems, clearly established benchmarks, and as yet that no algorithm shows neuromorphic systems outperforming deep learning in accuracy (Schuman et al. 2022).

Mehonic et al. discuss the use of memristors for in-memory computing, deep learning acceleration, and SNNs, emphasising the need for tailored learning and inference algorithms that exploit non-Von-Neumann architectures (Mehonic et al. 2020). Javanshir et al.’s review complements device-centric surveys by taking algorithmic and architectural perspectives of SNN algorithms, neuromorphic hardware platforms, and FPGA implementations (Javanshir et al. 2022).

Chen et al. review the promises and challenges of memristive devices in SNN implementations, focusing on diffusive memristors (used to mimic short-term plasticity (STP) due to their volatility) and drift memristors (used for long-term potentiation (LTP) during neural network training due to their stable retention) (P. Chen et al. 2025). They report that researchers developed fully memristive SNNs using diffusive memristors as stochastic LIF neurons and non-volatile memristors as synapses, achieving unsupervised learning for letter recognition.

A number of reviews of memristors emphasise analog tunability, endurance, and integration challenges, and point to neuromorphic computing as a key application area (Lee et al. 2022; B. Jin et al. 2025).

Threshold-switching memristors (TSMs) with volatile or semi-volatile dynamics naturally implement neuron-like behaviours such as oscillations, LIF spiking, and refractory periods through their intrinsic switching physics. Wang et al. review TSM-based spiking neuromorphic systems, cataloguing device materials and associated spiking modes, along with energy-per-spike figures (X. Wang et al. 2025).

2.3 Key Themes

2.3.1 Device-Level Memristor Synapses

Memristor synapses provide non-volatile conductance states with multi-level programmability, enabling in-memory computing and synaptic plasticity in SNNs (Prodromakis and Toumazou 2010). Systems use crossbar arrays of memristors to store synaptic weights (Kim et al. 2012; Wijesinghe et al. 2018; Bayat et al. 2018) and CMOS neuron circuits to implement spiking dynamics (Payvand et al. 2020).

Jin et al. provide a broader review of memristor-based networks for neuromorphic computing, covering convolutional, recurrent, and spiking networks, and emphasising memristor in-memory computing as a way to overcome the Von-Neumann bottleneck (B. Jin et al. 2025). Their discussion of synaptic plasticity mechanisms (potentiation, depression, and emerging learning algorithms tailored to memristive dynamics) is relevant to hardware-aware learning rules in SNNs, especially when device non-idealities constrain achievable weight updates.

Chandrasekaran et al. present a device-level critical review of memristor synapses, combining a statistical survey of neuromorphic device publications (2018-2025) with a discussion of biomimetic properties such as STP, LTP, STDP, and SRDP in various material systems (Chandrasekaran et al. 2026). They highlight optical-stimulus-driven 2D-material synapses and applications ranging from MNIST recognition to biomedical interfaces, and show the variety of physical mechanisms that allow synaptic plasticity.

2.3.2 Device-Level Ferroelectric and Antiferroelectric Synapses

Ferroelectric materials allow high-density, energy-efficient synaptic matrices with short-term and long-term memory effects. Park et al. report a 24x24 ferroelectric memristor crossbar array based on a HaO structure that demonstrates scalable device characteristics and neuromorphic abilities such as paired-pulse facilitation and spike-duration or frequency-dependent plasticity (Y. Park et al. 2025). They also use the ferroelectric array as a reservoir layer,

achieving high accuracy on Fashion-MNIST (more challenging than MNIST). The reported device-to-device and cycle-to-cycle uniformity addresses a challenge in memristive arrays, making these ferroelectric synapses attractive for hardware-aware learning, where variability and drift must be controlled.

Lu et al. review ferroelectric-based LIF neurons and outline device design and performance optimisation strategies, showing how leakiness is achieved by depolarization or antiferroelectric phases (Lu et al. 2026). They identify key metrics (hardware cost, energy consumption, endurance and application examples) and highlight co-design opportunities between devices implementing LIF and those implementing synaptic weight storage.

2.3.3 Device-Level Memristor and Ferroelectric Neurons

Earlier silicon neuron work such as the classical Mahowald and Douglas “A Silicon Neuron” (Mahowald and Douglas 1991) provide historical context and a baseline CMOS neuron architecture that current memristive and ferroelectric neuron designs build upon. But the area and static energy of capacitors and peripheral devices has created a recent shift towards device-intrinsic capacitance and polarisation dynamics in ferroelectric neurons.

Wu et al. present a reconfigurable LIF neuron with a single op-amp that accommodates a large number of memristor synapses and enables online STDP learning with optimised power consumption (Xinyu Wu et al. 2015). Their 180nm CMOS design is an early example of dense memristor-synapses and achieves high power-efficiency for STDP in large synapse populations with low integration current.

In 2018, Zhang et al. find that neuron implementations have lagged behind synapse implementations, because sophisticated neural dynamics have not been addressed. They demonstrate essential neural features (LIF, threshold spiking and self-recovery) in a single device (Yishu Zhang et al. 2018).

A series of recent works demonstrate compact artificial neurons implemented directly in memristive or ferroelectric devices, avoiding large capacitors and complex reset circuits while achieving rich spiking dynamics. Cao et al. fabricate a compact neuron based on an antiferroelectric field-effect transistor (AFFeT) using HfZrO₂, where intrinsic polarisation and spontaneous depolarisation implement LIF behaviour with low spike energy, high endurance, and strong uniformity (Cao et al. 2022). They model a two-layer SNN combining AFFeT neurons with ferroelectric synapses, reaching high MNIST accuracy, although this is only a simulation.

Chen et al. provide an overview of diffusive memristor neurons that emulate integrate-and-fire dynamics through field-accelerated drift and diffusion of metal atoms, and a perceptron network of two separated passive memristor crossbars (Chen et al. 2023). Zhang et al. fabricate a compact artificial memristive neuron with leaky integration, automatic threshold firing, and self-recovery in a single device with ultralow energy consumption, providing a device-level LIF neuron building block (Zhang et al. 2025).

Zhao et al. build a physically motivated model for a spiking neuron based on one diffusive memristor, one transistor, and one resistor, although the accuracy of a SNN based on this neuron was only determined in a simulation (Zhao et al. 2025).

Zhidkov et al. show multiple spiking modes (time-to-first-spike, spike count, firing-rate coding) in annealing-optimised Ag/Hf/Zr/O memristive neurons, realised using filamentary switching and a simple series resistor, illustrating how fabrication processes can tune spiking behaviour (Zhidkov et al. 2026). Wang et al. analyse threshold-switching memristors and review their use as spiking neurons with oscillatory, LIF, Hodgkin-Huxley-like, and stochastic dynamics, directly linking device physics (Mott vs. redox switching) to neuron models and energy consumption (X. Wang et al. 2025).

Gao et al. fabricate antiferroelectric transistors for reconfigurable neuromorphic functions via coupled polarisation switching and charge trapping, enabling multiple neuromorphic behaviours (e.g., synaptic and neuronal states) in the same device (Gao et al. 2025). Shi et al. fabricate a reservoir computing system with coplanar floating-gate antiferroelectric transistors that bring together volatile neuron-like spiking, nonvolatile synaptic behaviour, and fading-memory reservoir dynamics in a single device architecture (Shi et al. 2026).

Li et al. fabricate BiSe-based threshold-switching memristors with 10e6 switching ratios, low threshold voltages, and thermoelectric response, and use a HSPICE simulation to explore spiking responses to temperature variations (Li et al. 2025). This work shows how TSM neurons can be integrated with sensor capabilities in hardware, and how compact device models capture combined electrical and thermal dynamics for edge sensing.

Wang et al. fabricate a neuron device using a Pt/Ag/ZnO/Pt volatile diffusive memristor that exhibits threshold switching characteristics. They use diffusive memristors to produce LIF functions with low energy consumption,

and integrate these neurons into a two-layer spiking neural network for MNIST classification, achieving an 89.17% accuracy (L. Wang et al. 2025).

2.3.4 Multilayer Neuromorphic Perceptron Architectures

Passive memristive crossbars (0T1R) and active 1T1R arrays underpin many in-memory computing architectures for ANNs and SNNs, providing high-density weighted-sum operations and local update paths. Bayat et al. demonstrate a one-hidden-layer perceptron classifier implemented entirely in mixed-signal hardware using two passive 20x20 metal-oxide memristive crossbars, with off-chip training and a classification fidelity close to simulations despite device non-idealities (Bayat et al. 2018).

Tang et al. design (but only implement in FPGA) a fully memristive spiking-neuron learning framework built only from one drift and one diffusion memristive model, which eliminates the need for other electronic devices like resistors or capacitors. They apply the framework to a feedforward multilayer perceptron, to demonstrate its simplicity and high processing speed on hardware (Tang et al. 2020).

Poddar et al. report ultra-high density perovskite nanowire array memristors for multi-layer perceptron networks, emphasising nanowire crossbars as a path to extremely compact synaptic fabrics (Poddar et al. 2026).

A recent hybrid convolutional SNN architecture by Go and Kim combines digital non-leaky integrate-and-fire neurons with commercially-available Knowm memristor-based synapses in a 1T1R crossbar, using a current-sense amplifier and 1-bit comparator as a minimal readout instead of high-resolution ADCs (Go and Kim 2026). They use a temporal encoding scheme to reduce spike counts and device stress, and they highlight the energy usage of this minimalist readout versus conventional ADC-based designs and its importance for edge AI.

2.3.5 Learning Rules and In-Situ Training in SNNs

Biological and bio-inspired plasticity rules such as spike timing dependent plasticity (STDP), Hebbian learning, and voltage-dependent synaptic plasticity (VDSP) are central to in-situ learning in SNNs for local learning using device physics.

Diehl and Cook were first to show unsupervised digit recognition using STDP, lateral inhibition, and adaptive thresholds without any labels or teacher signals, achieving high MNIST accuracy and showing the robustness of biologically plausible mechanisms across different learning rules (Diehl and Cook 2015). Published memristor-CMOS neuromorphic chips have since demonstrated in-situ STDP learning in synapses (Karamimanesh et al. 2026), basic pattern recognition (Tavanaei and Maida 2018), and small-scale neural cores (Valentian et al. 2019).

Goupy et al. introduce Stabilized Supervised STDP (S2-STDP) combined with Paired Competing Neurons (PCN) for classification layers in SNNs, integrating error-modulated weight updates with local spike timing and intra-class competition (Goupy et al. 2024). They show improved performance over previous supervised STDP rules across multiple vision datasets, and their approach is suitable for hardware implementation due to its locality and lack of global backpropagation.

Recent work on Hebbian enrichment of SNNs proposes memory-augmented architectures where Hebbian synaptic plasticity acts as both memory and processor, improving out-of-distribution generalisation, one-shot learning, and reward-based learning (Limbacher et al. 2025). These Hebbian-enriched SNNs underline the computational versatility that can be achieved when synaptic updates are tightly coupled to local spike statistics, which is attractive for analog in-situ learning with memristive devices.

At the hardware level, energy- and endurance-aware co-design for edge AI highlights the need to design learning algorithms for device constraints such as limited endurance, stochastic switching, and restricted dynamic ranges (Go and Kim 2026). Hybrid schemes split feature extraction (often unsupervised STDP) from supervised classification layers and minimise write operations to demonstrate hardware-sympathetic learning strategies (Kugele et al. 2021).

2.3.6 Neuromorphic Reservoirs and All-in-One Architectures

Reservoir computing offers a flexible framework for temporal information processing using fixed recurrent substrates and trained readouts, which maps well onto analog device arrays with rich internal dynamics (Tan and Van Dijken 2023). See previously mentioned work on ferroelectric reservoir implementations (Y. Park et al. 2025; Shi et al. 2026).

2.3.7 Neuromorphic Edge and Embedded Applications

Several works in the bibliography focus on SNNs as controllers or embedded intelligence for resource-constrained platforms, such as FPGA-based neuromorphic systems and spiking neurocontrollers for UAVs. Qiu et al. evolve spiking neurocontrollers for full hexacopter control in six degrees of freedom, using modular decomposition and NEAT-style evolution, highlighting how spiking networks can be tailored to continuous control tasks (Qiu et al. 2020).

Kim et al.’s self-powered multimodal system integrates multimodal sensing, spike encoding, and unsupervised learning using drift and diffusive memristors, operating without digital circuitry or external power. It uses a SNN that directly receives multiple analogue inputs from physical sensors and converts these inputs into spike-like signals, which then modulate synaptic weights based on relative event timing (Kim et al. 2026).

2.4 Key Challenges Indicated by Literature

- **Device variability and imperfections** in memristive/analog crossbars cause weight noise, conductance drift, and cycle-to-cycle fluctuations that degrade inference accuracy and complicate training.
- **Non-ideal memristor dynamics** (non-linear weight updates, asymmetric potentiation/depression, retention loss, endurance limits) break assumptions of backpropagation and STDP.
- **Non-differentiable spiking** require surrogate gradients (SG) or ANN-to-SNN conversion; SG introduces approximation bias, memory-intensive BPTT, and poor biological plausibility.
- **Biological plausibility vs. engineering utility**: e.g. STDP is scientifically interesting but underperforms SG.
- **No consensus on computational primitives**: rate vs. temporal vs. population coding; integrate-and-fire vs. Izhikevich vs. Hodgkin-Huxley; local vs. global learning.
- **Generalization gap**: SNNs perform on temporal or sparse tasks but lack demonstrated reasoning, language, or planning capabilities of transformer-scale digital NNs.
- **No killer application** justifying neuromorphic large-scale production cost (>\$50M); edge AI applications served by low-power Microcontroller and Neural Processing Units, e.g. Texas Instruments TinyEngine NPU.

2.5 Synthesis and Gaps Relative to the Target Topic

The bibliography shows strong activity in device-level memristive and ferroelectric neurons and synapses, ferroelectric LIF neuron design, threshold-switching memristor-based spiking dynamics, and mixed-signal memristive crossbar demonstrators. It also includes several hardware-aware learning schemes; unsupervised STDP, supervised STDP variants, Hebbian-driven networks, and endurance-aware co-design, but these are studied either in simulation or with limited in-situ learning in small arrays.

Compact device models are commonly used in circuit-level simulations (e.g., HSPICE models, compact neuron models for memristive and ferroelectric devices), yet there remains a gap in unified compact models that simultaneously capture neuron dynamics, synaptic plasticity, and non-idealities for multi-layer analog SNNs. Fabricated demonstrations often focus either on device-level proof-of-concept (single neurons, small crossbars, reservoir layers) or on mixed-signal systems with off-chip training; fully analog, multi-layer SNNs with hardware-aware in-situ learning remain comparatively underexplored.

The literature indicates that key open problems include handling device variability and stochasticity, designing plasticity rules compatible with endurance limits, integrating memristive devices in multi-layer analog SNNs, and demonstrating small yet complete hardware prototypes that close the loop between learning, device physics, and circuit behaviour.

2.5.1 Gaps and Originality

There is a clear gap in scalable multilayer memristor SNNs with in-situ learning, as existing memristor implementations tend to focus on single-layer cores or small networks.

The proposed research, “hardware-aware in-situ learning in an analog SNN using measured memristive synapses and neurons, with a compact device model and a small fabricated demonstrator”, naturally builds on earlier work by combining measured device characteristics, compact unified models, and in-situ learning rules tuned to device constraints.

The significance of the research is that if in-situ learning for SNN can be shown to accommodate or even benefit from real-world device variability, even though the networks are still simple, the low-power design could benefit edge AI immediately, and scale to larger AI models with further research.

2.6 Research Objectives and Questions

2.6.1 Success Criteria

The project will be successful if it demonstrates:

- Physical credibility: Learning-rule simulations using empirical distributions from physical devices, not ideal device behaviour.
- Mechanism insight: Whether a device characteristic helps or harms learning, and how circuit or algorithm design can mitigate it.
- Practical evidence: at least one experiment updates physical states in response to spike-derived signals and improves a measurable task objective.

2.6.2 RQ1 - CMOS-Compatible Scalable SNN Architecture

What memristor-CMOS co-designs for synapses and neurons enable a scalable, multilayer spiking neural network that can be fabricated using the CMOS processes available at Zepler? Within this constraint, which crossbar topologies, neuron circuit models (e.g. LIF variants), and spike-routing schemes jointly optimise trainability, energy per spike, and silicon area for edge-scale event-based workloads?

2.6.3 RQ2 - Efficient In-situ Learning for Multilayer Memristor SNNs

Which in-situ learning algorithms-such as STDP/VDSP and hardware-feasible approximations to spiking backpropagation can be implemented efficiently in multilayer memristor SNNs, and how do their accuracy, convergence speed, and energy per update compare to ex-situ training with weight transfer under realistic device non-idealities?

2.6.4 RQ3 - Robustness and Exploitation of Device Variability

What levels of device-to-device and cycle-to-cycle variability in memristor synapses and neurons can a multilayer SNN tolerate before task accuracy degrades beyond a defined threshold, and how can in-situ learning rules be designed so that such variability either remains harmless or actively improves training speed and final accuracy?

2.6.5 RQ4 - System-level Performance on Event-Based Tasks

Can a multilayer memristor SNN with in-situ learning achieve competitive task accuracy, energy per event, and latency on representative event-based vision or time-series benchmarks compared with state-of-the-art software ANN/SNN baselines, when implemented under Zepler-compatible CMOS and realistic memristor non-idealities?

3 Proposed Methodology

3.1 Methodological Overview

The research will follow a two-stream approach, logical and physical, and both will progress from a foundation to greater degrees of sophistication. The logical stream will front-run the physical one, and will inform it, so that physical research (which generally has higher costs and longer leadtimes) is more likely to succeed.

The overall path is:

- physical device characterisation
- for simple network: model in software using device data -> apply learning rules in software -> construct simple physical network -> apply learning rules in hardware.
- for more complex network: model in software -> apply learning rules in software -> construct complex physical network -> apply learning rules in hardware .

3.1.1 Diffusive and Drift Memristor Characterisation

Use existing fabricated diffusive and drive memristors to characterise device behaviour, with a particular focus on device-to-device, cycle-to-cycle and stochastic variability.

Characterise fabricated devices and arrays to obtain IV curves, switching behaviour, endurance, retention, and variability, then fit compact models suitable for both circuit and network-level simulation.

Define an allowable variability specification.

3.1.2 Device and Circuit Design

Implement variability model in software emulation of a single, simple neuron. Convert memristor variability into neuron behaviour variability.

Use existing memristor expertise to select, design and fabricate CMOS-compatible memristor stacks with suitable characteristics (tuning, endurance, variability, and process compatibility).

Design hybrid synapse cells (e.g. 1S1R) and neuron circuits (e.g. LIF 1M1T1R neurons) that operate at realistic voltage ranges and support pulse-based conductance updates implementing STDP/VDSP or supervised learning rules.

3.1.3 Compact Modelling and Multi-scale Simulation

Use SPICE/Verilog-A simulation to validate synapse and neuron behaviour, including parasitic effects and sneak paths.

Move to a larger network, where feature extraction and differentiation becomes important. Understand whether variability assists in feature / node differentiation is assisted/accelerated by variability, or whether there is a point at which excessive variability becomes catastrophic for network performance.

Integrate these models into a multi-scale simulation stack linking device-level dynamics to spiking network behaviour, using Python SNN frameworks for training and analysis.

3.1.4 In-Situ Learning Schemes

Implement simulation of in-situ learning (STDP and VDSP) in software, for single, simple neurons (pre/post-synaptic), initially without variability. Introduce the variability statistical model created earlier.

Evaluate the impact of variability on learning speed and accuracy.

Implement unsupervised STDP/VDSP-based learning using local timing and pulse protocols that update memristor conductances based on spike correlations and presynaptic potentials.

Explore supervised learning schemes adapted to hardware, such as approximate backpropagation in the spiking domain, where error signals are encoded as additional spike trains or weight-update pulses.

Compare ex-situ training (software SNN trained first, then weights programmed into crossbars) to in-situ training (updates performed directly in hardware) under realistic device constraints.

3.2 Data, Resources, and Infrastructure

- ECS Zepler 4th floor measurement lab: MEMS probe station for DC and pulse tests of individual devices.
- Breadboard / pulse generator / oscilloscope / packaged devices: simple circuit testing.
- Zepler Nano Fabrication Lab: photo lithography, ALD thin-film deposition, lift off, annealing, etc.

The project is very dependent on nano-fabrication, but conversely, a physical implementation gives an advantage over other research which is limited to simulation.

3.3 Evaluation Strategy

Evaluate accuracy, energy per event, latency, and robustness on chosen tasks; compare across SNN hardware and software baselines using consistent metrics and datasets.

Perform sensitivity analyses to quantify the impact of device variability, limited conductance precision, and learning rule parameters on long-term performance and stability.

3.4 Risks, Assumptions, and Limitations

3.4.1 Risks

- **Fabrication available** A risk that although fabrication is required, timely training on necessary tools is not available due to trainer unavailability, or that training is ineffective and fabrication is incompetent and skills remain insufficient, or that fabrication tools are overbooked or suffer failure and downtime. Mitigation: Wherever possible, don't fabricate: use existing devices first. Consider outsourcing fabrication to ECS Partners (Zepler) or Von Ardenne (industry partner with fabrication capabilities), or device loan from other research institutions.
- **Devices unsuitable** A risk that existing volatile memristor designs for which Zepler has fabrication experience are not suitable to model neurons, either because the device-to-device variability is too high, the spiking behaviour is incorrect, or the endurance is poor. Mitigation: Consider alternative neuron implementations, either using different materials e.g. HfO, or different behaviours e.g. AFeFET.
- **Crossbar effects** A risk that devices work individually, but not when fabricated into a dense array, e.g. crossbar synapses, due to unmanageable side-effects such as sneak-path currents. This risk field is well understood, but optimum solutions like 1S1R are still the subject of research. Mitigation: Consider alternative, more established management approaches, e.g. 1T1R, even if suboptimal from a space/energy perspective.
- **In-situ learning** A risk that in-situ learning approaches e.g. STDP don't translate to hardware without a large accuracy drop, due to device non-linearity and variability. Mitigation: This is one of the research goals, so absolute mitigation is not required, but an advanced indication can be provided through effective, software simulation of devices and learning approaches using detailed device behaviour from careful characterisation and measurement.

3.4.2 Assumptions

Fabrication budget and training Assumption that fabrication is possible and practical, that budget exists and training will be received. Validation: previous ECS PhD students have followed this path and achieved research within established budgets.

3.5 Skills Audit Training Needs

Skill / capability gap	Why it matters	Planned training or action	Indicative timing
Device measurement	Needed to characterise devices for software simulation	B53 measurement lab induction, training	Sept 2026
Device fabrication	Needed to fabricate devices for physical evaluation	B53 nano fab lab induction, tool training	Oct-Dec 2026
Foundational understanding of nano-electronics	Underpins memristor behaviour	Audit ELEC3207 or ELEC2312 module	Semester 1, 2026
Software development	Needed to assess and develop SNN libraries	None required	N/A

4 Impact Assessment

4.1 Overview of Anticipated Impact

This impact assessment evaluates the societal, ethical, and environmental implications of developing an analog spiking neural network (SNN) that integrates measured memristive synapses and neurons with a compact device model for hardware-aware in-situ learning.

By 2030, AI global data centres using digital hardware are projected to consume around 945 TWh of electricity annually, while their associated water footprint could equal the basic domestic water needs of 1.3 billion people and their land footprint could exceed 14,500 km² (De Vries-Gao 2026), loading power grids, competing with households and agriculture for scarce water, and consuming land for facilities, power generation and transmission networks.

These impacts are unevenly distributed: communities hosting data centres may face higher utility costs, local water stress, noise and construction impacts, while the benefits of AI accrue disproportionately to technology companies and wealthier users. The problem is intensified by everyday inference and deployment, which the UN University analysis estimates account for roughly 80-90% of total AI energy demand, rather than training alone (Aczel et al. 2026). Consequently, without limits on efficiency, siting, disclosure and resource use, AI development could deepen environmental inequality by shifting the ecological costs of digital services onto regions least able to absorb them.

By co-locating memory and processing into event-driven analog memristive crossbars, this research can address the “Von-Neumann bottleneck”, offering energy efficiency orders of magnitude higher than conventional digital hardware (GPUs/TPUs). Spiking neural networks process information sparsely and can potentially reduce computation, data movement and network traffic in applications such as continuous sensing and environmental monitoring. This could lower operational electricity demand and reliance on cloud infrastructure, and indirectly reduce associated cooling-water and data-centre land requirements. However, neuromorphic hardware manufacturing, preprocessing, communications and end-of-life impacts must also be included, and like-for-like measurements are needed before a net life-cycle advantage can be claimed (Aral 2024).

Additionally, making the transition of analog neuromorphic computing from lab prototypes to production deployment introduces complex trade-offs across sustainability, reliability, and governance. The benefit of memristive neuromorphic systems will depend as much on robust RRI governance, standardized testing, amelioration of device drift and variability, and sustainable material supply chains as it does on raw power efficiency gains.

4.2 Considerations Under the 4Ps Framework

The 4P dimensions cover Process, Product, Purpose and People. Considering these in turn:

4.2.1 Process

The research is carried out predominantly in the Zepler cleanrooms of the university. The focus will be on developing a documented and repeatable device measurement and fabrication process, with minimum waste of rare earth minerals and metals, and devices fabricated only at the point of need (no stock waste). Established recipes will be used to maximise yield. There aren’t considered to be any significant data management risks or ethical issues, and consent is not required. Widely-adopted test datasets will be used, which contain no personal information. The standard Zepler governance process and rules will be followed at all times. Lab safety is an extremely important consideration, since many of the fabrication chemicals can be harmful, so a full programme of safety induction and specific process training will be followed.

In-situ learning enables local adaptation directly on edge hardware without transmitting raw sensor streams (video, audio, biomedical signals) to centralized servers. This “privacy-by-default” architecture aligns with strict data protection regulations (e.g., GDPR) by keeping sensitive personal data local and volatile.

4.2.2 Product

The research outputs (algorithms, devices, platforms, datasets, papers) will be made public, with test specifications and results available through a Creative Commons license to enable reproducibility of experiments. A successful conclusion of research will lead to working prototypes at a low Technology Readiness level (likely TRL 3: Analytical and laboratory proof-of-concept tests are completed). A prototype may be capable of simple classification and other machine learning tasks, but will not outperform the accuracy of conventional digital artificial neural networks, and will not be directly usable.

Longer term, it is anticipated that the technology may deliver AI to low-power edge devices, and this should be put to beneficial uses, such as environmental monitoring, but also undesirable ones, such as drone warfare. Low-power in-situ intelligence significantly expands the capabilities of untethered autonomous systems, including micro-drones and target-tracking sensors. The low thermal and power signature of memristive SNNs makes them inherently attractive for defense systems, raising non-proliferation concerns and ethical challenges regarding Lethal Autonomous Weapons Systems (LAWS). It’s not possible to control the technology so that it could only be used in positive

use-cases. Another consideration is the ability to audit or constrain behaviour, but neuromorphic systems face the same challenges as digital AI counterparts.

4.2.3 Purpose

The research will align with the public-interest goals of improving health and increasing sustainability of human activity. The main driver is to seek a plausible reduction in the energy consumption of AI training and inference, and a consequent consumption of cooling water, although neuromorphic computing is as a whole at a very low readiness, and so the contribution may not emerge for a decade or more.

4.2.4 People

The primary impact on people for the duration of this project is on researchers. It’s premature to consider end-users or affected communities in detail (e.g. workers displaced by more capable automation, or people subject to always-on sensing), since the technology is at a very early readiness level, but the intention is to benefit communities through alignment with sustainable development goals.

Device physicists & chip designers Responsible for characterization accuracy and compact model fidelity; must prioritize reproducible physical metrics over optimistic simulation benchmarks.

Semiconductor foundries Require standardized back-end-of-line (BEOL) integration processes to ensure scalable manufacturing without introducing heavy-metal contamination to cleanrooms.

Edge application developers Benefit from localized learning capabilities; must design compensations against analog drift and handle non-deterministic hardware behaviors.

Regulatory bodies & policy makers Need to establish safety standards for analog/neuromorphic AI chips, ensuring hardware calibration and accountability in safety-critical deployments. Need to help policy makers (e.g. politicians) understand the relevance of low-power AI in an era of increasing public scepticism of the environmental impact of AI.

4.3 Considerations Under the AREA Framework

4.3.1 Anticipate

The project proactively models hardware non-idealities-such as cycle-to-cycle conductance variation, random telegraph noise, and thermal drift, before physical scale-up. It is anticipated that device-to-device variability may lead to slow progress, and other neuromorphic research projects have remained in the examination of single devices rather than complete systems. To mitigate this, the project will use established recipes and work with what exists, rather than optimise the devices themselves.

Memristive devices exhibit non-volatile state degradation over extended write cycles. If an analog SNN adapts continuously in-situ without strict stability bounds, device degradation could lead to unpredictable weight shifts. In safety-critical applications like bio-implantable medical monitors or driver-assist sensors, uncalibrated drift introduces severe safety hazards.

4.3.2 Reflect

The project must reflect on the hype cycle surrounding neuromorphic engineering. Demonstrations on “small fabricated prototypes” must be evaluated critically regarding scalable yield, CMOS compatibility, and realistic manufacturing costs rather than assuming frictionless commercial adoption. Sustainability is embedded in the project, with a total focus on reducing the energy, water and land consumption of AI, and the associated carbon emission and climate change impacts.

Analog SNNs offer transformative reductions in operational energy consumption, but their total environmental footprint includes embodied emissions and material toxicity from advanced fabrication.

4.3.3 Engage

Development requires cross-disciplinary dialogue between material scientists, laboratory technicians, computer scientists, and domain-specific end users (e.g., medical device designers, environmental sensor operators) to align hardware constraints with human safety requirements.

To make neuromorphic computing a reality and advance it to the point of societal benefit, further funding will be required. Engagement with funding authorities such as UKRI and policy makers such as UK parliamentary committees is needed, at all times demonstrating that research monies are being spend responsibly.

Outreach activities are being planned, such as the British Science Festival in Southampton in 2027, where the concepts of modelling computers on the human brain could be demonstrated (such as a LED-powered synapse, or a water-level neuron trigger model) to inspire future scientists.

4.3.4 Act

Hardware-aware models must be adapted as fabrication of alternative material stacks (e.g., transition from HfO to TaO layers) becomes possible, or other devices (e.g. Anti Ferroelectric transistor devices for neurons) are proven, ensuring algorithms remain grounded in empirical physical behavior.

4.4 Relevant Sustainable Development Goals

UN SDG	Title	Direct Impact	Key Mechanisms & considerations
SDG 7	Affordable & Clean Energy	High Positive	Cuts compute energy by 10x to 100x via sparse, event-driven spike processing and local analog weight updates, reducing grid strain from AI workloads.
SDG 9	Industry, Innovation & Infrastructure	High Positive	Enables energy-harvesting edge IoT devices, autonomous sensors, and decentralized infrastructure monitoring without relying on power-hungry cloud data centers.
SDG 12	Responsible Consumption	Moderate Negative / Mixed	Memristor fabrication often utilizes novel transition metals (Hafnium, Tantalum, Titanium). Scaling requires strict e-waste lifecycle management and sustainable chemical etching protocols.
SDG 13	Climate Action	Positive	Mitigates the growing global carbon footprint of artificial intelligence training and continuous edge inference ecosystems.

Overall, responsible research and innovation (RRI) principles are being considered to ensure that technological advances in neuromorphic systems align with societal values and long-term risk management.

5 PhD Plan

5.1 Overview of Direction

The overall trajectory of the PhD is to first measure and understand existing device behaviour, then construct simple end-to-end systems from them, then understand how local in-situ learning rules are affected by device variability.

The approach is to reuse known device structures and fabrication recipes (supervisory knowledge), and NOT to experiment with novel fabrication layers or materials per se. Develop pulse characterisation, automation, data schemata, analysis, and circuit test groundwork should be completed before fabrication.

In an ideal world, the learning rule evaluation should direct what devices are required, rather than the alternative (in which the availability of devices drives the rule evaluation).

5.2 First 9 Months

5.2.1 Goals:

- Characterise pre-made drift (synapse) and diffusion (neuron) memristors using MEMS probe station. I/V DC tests, set / reset behaviour, pulse tests, spiking behaviour, device-to-device variability.
- Define pulse specification for spiking dataset, e.g. N-MNIST.
- Construct multilayer spiking neural network software model for laptop execution; 60,000 test samples, 5,000 events per sample. Model first level of device behaviour; e.g. spiking charge threshold, refractory behaviour, minimum and maximum conductance.
- Device material choice, specification and recipe selection for fabrication of synapse and neuron devices.
- Device packaging for breadboard deployment.
- Design and construction of simple 2-synapse, 1-neuron model on breadboard.
- Evaluation of in-situ learning rules (spike timing dependent plasticity, voltage dependent synaptic plasticity).

5.2.2 Detailed Plan

Month	Planned activities	Deliverables / milestones
1	Complete formal literature review on memristor neuromorphic devices, in-situ learning in SNNs	Review
1-2	Measurement training (MEMS, Pulse, Cascade)	Ability to characterise devices
4-6	Nano fab training	Ability to fabricate devices
3	Poster / participation at Neumat London November 2026	Poster
2-5	Characterise synapse and neuron devices	Library of measurement settings and methods, and device behaviour data
5	Internal presentation on Spiking Neural Networks to SustAI cohort	Presentation
3-6	Build software simulation of 2 synapses / 1 neuron, incorporating actual device variability dimensions and data	Data for simulated behaviour of system
6-9	Evaluate in-situ learning methods, showing device variation impact	Learning performance (accuracy, rates, energy consumption)
6	Poster / participation at Neumat Winter School 2027	Poster
8	SustAI conference	Presentation
9	Preparation for 9-month progression review	Review paper

5.3 Year 1 Month 10-12 Goals

- Extend 9-month work to simulation of multilayer network, in order to assess learning performance (accuracy, rates, energy consumption)
- Complete all cleanroom training required for circuit fabrication.
- Southampton Science Festival 2027
- A publishable simulation/circuit paper that does not depend on successful fabrication (de-risk).

5.4 Year 2 Goals

- Selection of crossbar array design to reduce sneak path currents (e.g. 1S1R, 1T1R etc.).
- Design and simulation of larger, multilayer memristor synapse and neuron circuits, including layout of small crossbar synapse arrays.
- Evaluation of alternative in-situ learning approaches (e.g. STDP, VDSP etc.).
- Fabrication of test structures at Zepler.
- Optional exploratory work on AFET-based synapse/neuron concepts as future extensions.
- A publishable measured-device validation paper.

5.5 Year 3 Goals

- Further development of in-situ learning schemes (STDP, VDSP and supervised variants) in the network.
- Demonstration of learning on small edge-AI task aligned with sustainability, e.g. environmental sensing.
- Comparative evaluation of in-situ learning against software SNN baselines.
- Sensitivity and robustness studies; derivation of design guidelines and preparation of main publications.
- A publishable paper describing an integrated demonstrator.
- Consolidation of results and thesis write-up.

5.6 Gantt Chart

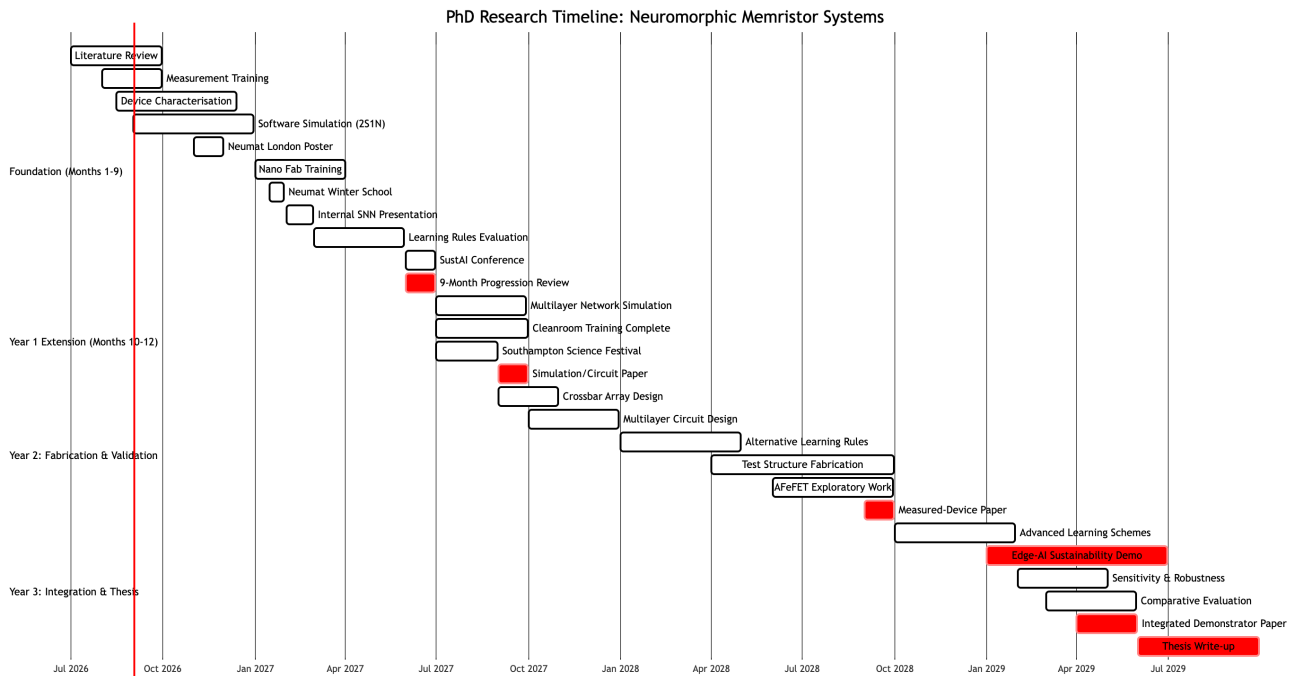


Figure 1: PhD Gantt Chart

5.7 Meeting and Supervision Plan

Weekly face-to-face meeting with Dr. Firman Simanjuntak and Prof. Mark Zwolinski

5.8 Contingency Planning

By March 2027, evaluate whether reliable access to a device dataset exists which could support a publishable hardware-aware learning result by March 2028. If so, proceed with integrated device-circuit-algorithm development; if not, pivot from fabrication to variability-aware learning rules.

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7 Appendices

7.1 Appendix A. Characterisation Detail.

7.1.1 Synaptic Behavior of Drift Memristors.

Test	Protocol	Extract for the model
1. Read linearity and state range	Read multiple programmed states using a low, non-disturbing voltage	Gmin_G_{\min}Gmin_, Gmax_G_{\max}Gmax_, read noise, read disturb, usable number of levels
2. Incremental potentiation	Start from a defined low-conductance state; apply identical SET or potentiation pulses, reading after every pulse	GGG-versus-pulse-number curve, update granularity, saturation, stochasticity
3. Incremental depression	Start from a defined high-conductance state; apply RESET or depression pulses, reading after every pulse	Depression curve, asymmetry relative to potentiation, saturation, stochasticity

Test	Protocol	Extract for the model
4. Pulse-amplitude/width dependence	Repeat selected updates for several amplitudes and widths	Update rule $dG(G,V,\tau)\backslash\Delta G(G,V,\tau)dG(G,V,\tau)$, programming threshold, energy per update
5. Retention/drift	Program representative low, middle, and high conductance states; read over logarithmically spaced delays with no programming bias	Drift law, retention distribution, state-dependent drift, effective weight lifetime
6. Cycle-to-cycle and device-variation test	Repeat potentiation/depression cycles on several devices	Parameter distributions, update covariance, stuck states, device mismatch
7. Endurance subset	Repeatedly alternate potentiation and depression while periodically checking the conductance window	Cycle-induced degradation, failure probability, changing update curves

7.1.1.1 Core Static Electrical Behaviour.

- I/V curves over expected read and write voltage ranges, including nonlinear conductance.
- Hysteresis under voltage sweeps.
- High-resistance and low-resistance limits.
- Conductance dynamic range and usable analogue levels.
- SET and RESET polarity.
- Threshold voltage or threshold field.
- Read disturbance caused by sub-threshold read pulses.
- Current compliance effects.
- Series resistance and parasitic capacitance because the device is to be simulated in a circuit-level setting.

7.1.1.2 Pulse Behaviour.

Single-pulse response:

- Peak conductance.
- Conductance decay.
- Delay between excitation and conductance change.
- Pulse-amplitude and pulse-width dependence.

Pulse-pair response:

- Paired-pulse facilitation.
- Paired-pulse depression.
- Dependence on pulse order.
- Dependence on positive and negative Δt .
- Effective STDP window.

Pulse trains:

- Temporal summation.
- Short-term facilitation and depression.
- Frequency-dependent potentiation.
- Frequency-dependent depression.
- Transition from volatile to persistent conductance change.
- Dependence on inter-pulse interval.
- Rate-to-conductance transfer function.

Drift:

- Conductance drift rate.
- Dependence of drift on the programmed state.
- Relaxation after SET and RESET separately.

- Temperature dependence.
- Retention distribution, not just mean retention.
- History dependence.
- Drift after repeated programming.
- Cycling-induced change in the drift law.

7.1.2 Neuronal Behaviour of Diffusion Memristors.

Test	Protocol	Extract for the model
1. DC or slow voltage sweep	Sweep voltage from 0 through the positive excitation range, then return to 0; repeat for several amplitudes and compliance limits	Threshold voltage, threshold current, hysteresis, ON-state conductance, switching polarity
2. Single-pulse spike test	Apply isolated voltage pulses with a grid of amplitudes and widths; return to zero after each pulse	Spike/no-spike probability, spike latency, peak current, spike width, energy, threshold surface $P_{\text{spike}}(V, \tau)$
3. Relaxation test	After a pulse that produces a conductive state, measure conductance/current at logarithmically spaced delays with zero bias	Volatile decay law, relaxation time, residual conductance, dependence on pulse amplitude and width
4. Pulse-pair interval test	Apply two identical pulses with inter-pulse interval Δt , varying Δt over the intended SNN range	Temporal summation, paired-pulse facilitation/depression, effective memory kernel
5. Repeated-pulse train test	Apply trains at several frequencies and amplitudes	Frequency-dependent firing, accumulation, burst threshold, spike-rate response, transition between isolated and sustained spiking
6. Repetition and device-variation test	Repeat tests 2-5 over many cycles and multiple devices	Cycle-to-cycle and device-to-device distributions of threshold, latency, spike width, relaxation, and failure modes

7.1.2.1 Core Electrical Behaviour.

- The magnitude of conductance change per pulse.
- Dependence on pulse amplitude, width, and polarity.
- Nonlinear potentiation and depression.
- Saturation near conductance boundaries.
- State-dependent update sensitivity.
- Incremental versus abrupt switching.
- Update asymmetry.
- Dependence on pulse history and pulse spacing.
- Dependence on compliance current.
- Recovery or relaxation between programming pulses.
- Conductance relaxation after a pulse.
- Relaxation time constant or distribution.
- Whether relaxation is exponential, stretched exponential, power-law, or multi-timescale.
- Dependence of relaxation on the peak conductance reached.
- Dependence on pulse duration and amplitude.
- Residual conductance after partial relaxation.
- Facilitation from closely spaced pulses.

- Transition from volatile to effectively nonvolatile behaviour at high excitation.

7.1.2.2 Diffusive Characteristics:

- Threshold for spike initiation.
- Spike latency versus input amplitude.
- Spike latency versus pulse number.
- Temporal integration window.
- Firing-rate versus input-current relationship.
- Interspike interval distribution.
- Refractory period or refractory-like suppression.
- Adaptation after repeated spikes.
- Burst firing.
- Oscillation regimes.
- Noise-induced spiking.
- Spike amplitude, width, and energy.
- Reset dynamics.
- Dependence on temperature and bias history.

7.1.2.3 Variability

- Device-to-device variation.
- Cycle-to-cycle variation.
- Initial-state variation.
- Variation in conductance min and max.
- Variation in threshold voltage.
- Variation in switching delay.
- Variation in relaxation time.
- Variation in potentiation/depression slopes.
- Correlation between parameters within one device.
- Spatial correlation across an array.
- Temporal ageing and drift.
- Stuck-ON and stuck-OFF failure modes.
- Read noise and programming noise.

7.1.2.4 Stochastic behaviour.

Stochastic behaviour is not likely to be simple Gaussian noise on conductance. Instead, it may depend upon:

- Switching delay.
- Filament nucleation probability.
- Filament rupture probability.
- Dwell time in metastable states.
- Cycle-to-cycle variation.
- Random telegraph switching (can be a major source of temporal variability and read/write uncertainty).
- Noise-mediated threshold crossing.
- Correlation between successive updates.